

MARGARET LAKE, YT, Canada

WMO: 719770

Lat: 68.801N

Lon: 140.847W

Elev: 1864

StdP: 13.73

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.2	-37.6	-48.4	0.3	-40.7	-44.8	0.4	-37.8	14.9	-2.4	12.3	-4.6	0.8	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	73.2	58.1	69.8	56.2	66.7	54.9	60.6	69.5	58.3	67.0	56.4	64.6	7.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.8	73.8	63.7	54.5	67.8	61.5	52.3	62.4	60.1	27.7	69.7	26.1	67.3	24.8	64.7	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.0	8.8	DB	-43.9	78.8	5.6	2.7	-48.0	80.7	-51.3	82.3	-54.4	83.9	-58.5	85.9	(4)
(5)			WB	-44.4	62.1	5.4	1.9	-48.3	63.5	-51.4	64.6	-54.5	65.7	-58.4	67.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	-15.0	-13.0	-5.5	11.1	35.5	50.8	53.8	47.3	34.1	12.3	-7.1	-13.7	(6)
(7)		DBStd	27.92	13.31	12.36	11.74	11.24	11.78	7.87	6.00	6.97	8.39	10.35	11.33	11.13	(7)
(8)		HDD50	12714	2015	1763	1722	1166	461	83	34	136	479	1167	1714	1974	(8)
(9)		HDD65	17867	2480	2183	2187	1616	914	428	347	549	928	1632	2164	2439	(9)
(10)		CDD50	325	0	0	0	0	12	107	153	53	0	0	0	0	(10)
(11)		CDD65	2	0	0	0	0	0	1	1	0	0	0	0	0	(11)
(12)		CDH74	64	0	0	0	0	0	31	23	10	0	0	0	0	(12)
(13)		CDH80	3	0	0	0	0	0	1	1	1	0	0	0	0	(13)
(14)	Wind	WSAvg	2.7	1.8	1.5	1.7	2.5	3.7	4.7	4.3	3.9	3.1	2.2	1.6	1.4	(14)
(15)	Precipitation	PrecAvg	9.8	0.5	0.4	0.4	0.3	0.5	1.2	1.8	1.8	1.1	0.9	0.6	0.5	(15)
(16)		PrecMax	12.3	1.3	0.9	0.9	0.8	1.3	2.4	3.5	3.6	2.2	1.5	1.1	1.1	(16)
(17)		PrecMin	7.5	0.1	0.1	0.0	0.0	0.1	0.3	0.7	0.5	0.6	0.4	0.1	0.1	(17)
(18)		PrecStd	1.3	0.3	0.2	0.2	0.2	0.2	0.6	0.5	0.6	0.4	0.3	0.3	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	20.6	28.9	43.3	70.8	78.2	76.6	75.1	58.7	38.3	21.4	17.0	(19)
(20)			MCWB	20.0	18.9	25.9	36.0	53.0	61.1	60.3	61.2	47.2	35.9	19.8	16.2	(20)
(21)		2%	DB	12.3	15.2	20.5	38.3	65.5	73.6	72.7	69.1	54.5	32.6	16.5	9.8	(21)
(22)			MCWB	11.7	14.0	18.3	32.9	51.0	58.6	57.8	56.7	45.8	31.1	15.4	9.0	(22)
(23)		5%	DB	8.5	9.1	15.8	34.0	59.6	69.6	69.5	64.5	50.6	29.7	13.0	5.9	(23)
(24)			MCWB	8.0	8.4	14.2	30.3	48.6	56.3	56.5	53.5	43.9	28.2	12.1	5.0	(24)
(25)		10%	DB	3.4	4.2	11.0	30.1	54.2	66.3	66.7	60.6	46.7	26.7	9.7	2.0	(25)
(26)			MCWB	2.9	3.5	9.5	26.6	45.7	54.4	55.7	51.8	42.3	25.1	8.8	1.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	19.6	26.6	36.8	54.8	64.8	62.7	62.6	49.3	36.2	19.7	16.1	(27)
(28)			MCDWB	22.0	20.5	29.0	41.9	67.0	75.3	72.8	74.1	55.7	37.8	21.7	17.0	(28)
(29)		2%	WB	11.5	14.5	18.8	33.5	52.1	60.9	60.4	58.0	46.6	31.4	15.7	8.9	(29)
(30)			MCDWB	11.9	15.3	20.8	37.6	63.6	69.2	68.8	67.2	51.9	32.3	16.6	9.5	(30)
(31)		5%	WB	7.5	8.9	14.1	30.3	49.7	57.9	58.6	55.2	44.5	28.3	12.2	5.1	(31)
(32)			MCDWB	8.1	9.5	15.8	34.3	59.7	67.2	66.7	63.2	49.9	29.6	13.0	6.0	(32)
(33)		10%	WB	2.0	3.8	9.5	26.7	46.3	55.7	56.9	52.6	42.4	25.0	8.9	1.5	(33)
(34)			MCDWB	2.5	4.4	10.8	30.3	54.4	64.3	64.8	59.5	46.6	26.4	9.7	2.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.4	14.4	18.6	22.2	20.7	21.2	19.1	18.7	15.5	13.4	12.5	13.3	(35)
(36)			MCDBR	18.7	18.1	20.5	24.0	27.8	26.3	24.8	26.0	22.6	14.1	15.2	17.9	(36)
(37)			MCWBR	18.6	17.3	16.5	18.7	16.0	12.9	12.4	13.7	14.2	12.3	14.0	17.7	(37)
(38)		5% WB	MCDBR	18.3	17.9	18.7	23.0	26.8	24.6	21.9	22.6	20.3	14.2	14.6	18.5	(38)
(39)			MCWBR	18.0	17.2	16.5	18.3	16.6	13.5	12.0	12.4	13.2	12.4	14.2	18.1	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.160	0.202	0.240	0.296	0.325	0.335	0.361	0.323	0.280	0.211	0.148	N/A	(40)
(41)		taud		1.902	2.147	2.246	2.149	2.237	2.417	2.379	2.494	2.490	2.185	1.918	N/A	(41)
(42)		Ebn at Noon		26	197	258	267	272	274	260	261	245	185	25	0	(42)
(43)		Edh at Noon		2	13	22	34	35	30	31	24	18	11	1	0	(43)
(44)	All-Sky Solar Radiation	RadAvg		8	151	690	1445	1892	1948	1578	1096	622	213	25	0	(44)
(45)		RadStd		1	12	34	59	125	195	115	102	68	24	2	0	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+1.57	+1.66	N/A	N/A	N/A	N/A	-567	-583	N/A	N/A

Nomenclature: See separate page